

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	20-08-2026	Duration:	60 minutes

Code of Conduct

I K. BHAVANA (192524178) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

K. Bhavana

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. **Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.**
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Case Study 33 – SDG 5:

Women Entrepreneurship Support Portal

Many women-owned small businesses struggle to access government schemes, financial assistance, and business mentoring. Existing information is fragmented across multiple websites. A government agency plans to create an integrated digital platform supporting registration, funding opportunities, training programs, and networking. As a software engineer, analyze the above problem and propose a suitable software engineering solution.

1. Introduction

United Nations Sustainable Development Goal 5 (Gender Equality) calls for equal access to economic resources, financial services and property ownership for women. Women-owned micro, small and medium enterprises frequently face difficulty in discovering and accessing government schemes, capital grants, capacity-building training and mentorship, because this support is dispersed across multiple, disconnected government websites and offices.

The Women Entrepreneurship Support Portal (Case Study 33) is a role-based, full-stack web application developed to address this problem by consolidating government schemes, funding opportunities, training programmes, mentor directories and networking events onto a single centralized platform. The system supports three actors — the Women Entrepreneur, the Certified Mentor, and the Government Officer — each with a dedicated dashboard and workflow.

This document is the CO4 Assessment Tool 1 deliverable and focuses exclusively on test case design for the above system. Its purpose is to systematically identify the functional and non-functional requirements to be verified, derive test scenarios, and design detailed, traceable test cases covering normal, boundary, invalid and exceptional conditions — in line with CO4: “Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics.”

Rigorous test design is essential for a platform of this nature because it directly handles sensitive business, financial and identity information (Udyam registration numbers, turnover, funding applications) for a vulnerable user segment; incorrect verification, funding, or access-control logic could deny a legitimate entrepreneur access to public support or expose one user's data to another.

2. Problem Statement

Case Study 33 – SDG 5: Women Entrepreneurship Support Portal states that many women-owned small businesses struggle to access government schemes, financial assistance, and business mentoring because existing information is fragmented across multiple websites. A government agency therefore plans to create an integrated digital platform supporting registration, funding opportunities, training programmes, and networking. As a software engineer, the objective is to analyse this problem and propose a suitable software engineering solution — which, for this assignment, has already been realised as the Women Entrepreneurship Support Portal described in the previous technical report, and is now the subject of test case design.

3. Proposed System Overview

The Women Entrepreneurship Support Portal is a centralized, role-based platform with three actors, each provided a dedicated dashboard:

- Women Entrepreneur: registers a business profile (Udyam number, sector, turnover, employee count), browses and searches government schemes and funding opportunities filtered by sector, applies for funding through a multi-step application form, tracks application status on a visual progress timeline, enrolls in training workshops, and requests mentorship from certified mentors.

- **Certified Mentor:** manages profile and availability, reviews and accepts/rejects mentorship requests, and conducts advisory sessions with meeting links and feedback capture.
- **Government Officer:** audits and verifies registered businesses (approve/reject with mandatory rejection remarks), manages the catalogue of schemes, funding opportunities, training programmes and networking events, reviews funding applications, manages announcements and user accounts, and generates filterable reports with CSV export.

Core system capabilities include: government scheme discovery, funding opportunity applications, training programme enrolment, networking events, mentorship workflows, business profile management, government (officer) verification, an Entrepreneur dashboard, and an Officer dashboard — all secured through role-based access control.

Technology Architecture

As documented in the previous technical report, the system follows a three-layer architecture:

- **Presentation Layer:** React 18 single-page application built with Vite, using React Router and Recharts.
- **Application & API Layer:** Node.js and Express.js REST API, with JWT-based authorization middleware restricting access by role.
- **Data Access Layer:** a WebAssembly-compiled SQLite engine (sql.js) with disk-synchronized persistence, comprising sixteen normalized relational tables.

In summary, the request flow is: React/Vite frontend → Node.js/Express REST API → SQLite (sql.js) data layer, deployed via Docker. (Note: the previous technical report documents the database engine as SQLite/sql.js rather than MySQL; this report follows the previous technical report as the authoritative source for system facts.)

4. Functional Requirements to be Tested

The functional requirements below are reproduced from the previous technical report's Requirements Analysis (Table 1) and restated here with a testing objective for each.

Req. ID	Functional Requirement	Testing Objective
FR1	Women Entrepreneur Registration	Verify that a new Women Entrepreneur / Certified Mentor can register via the multi-step registration form with valid and invalid data.
FR2	Secure Authentication	Verify that JWT-based login correctly authenticates valid users and rejects invalid credentials.
FR3	Entrepreneur Profile Management	Verify that personal details and address information can be created, updated and validated.
FR4	Business Information & Udyam Registration	Verify that business details (Udyam number, sector, turnover, employee count) are captured and validated correctly, including duplicate Udyam guard.
FR5	Government Schemes Catalogue	Verify that the centralized schemes directory is correctly displayed to entrepreneurs and officers.
FR6	Scheme Search & Filter	Verify keyword search and sector-based filtering of schemes return correct results.
FR7	Funding Opportunities	Verify that capital grant and soft-loan listings are correctly displayed.
FR8	Funding Applications	Verify that the multi-step funding application form validates input and persists the application.
FR9	Application Progress Tracking	Verify that the visual step timeline correctly reflects application status and officer remarks.
FR10	Training Programme Management	Verify capacity-building workshop listing and officer-side management.
FR11	Training Enrollment	Verify workshop registration, duplicate-enrollment guard and capacity guard.
FR12	Mentor Directory	Verify that certified mentors are correctly listed and searchable.

Req. ID	Functional Requirement	Testing Objective
FR13	Mentorship Requests	Verify the 1-on-1 mentorship request queue and mentor response workflow.
FR14	Mentorship Sessions	Verify meeting-link generation and feedback capture for mentorship sessions.
FR15	Networking Events	Verify that summits and networking meets are correctly listed.
FR16	Networking Registration	Verify digital event-pass registration and duplicate-registration guard.
FR17	Announcement Management	Verify ministry notice publishing and display in the entrepreneur feed.
FR18	In-App Notifications	Verify real-time notification delivery through the navbar notification drawer.
FR19	Entrepreneur Verification	Verify the officer-side business verification workflow, including mandatory rejection remarks.
FR20	Reports & Analytics	Verify filterable query reports and CSV export for government officers.
FR21	Dashboards & Statistics	Verify role-specific dashboards with correct statistics and chart visualizations.
FR22	User Management	Verify officer-side account inspection and status control.

5. Non-Functional Requirements to be Tested

The non-functional requirements below are reproduced from the previous technical report's Requirements Analysis (Table 2), restated with a testing objective for each.

Req. ID	Non-Functional Requirement	Testing Objective
NFR1	Security	Verify password hashing (bcrypt), JWT session security and protection against unauthorized access.
NFR2	Performance	Verify acceptable response times for key operations and correct loading of the minified production build.
NFR3	Usability	Verify consistency of colour palette, status badges, navigation and validation messages across dashboards.
NFR4	Reliability	Verify correct behaviour and data persistence when the backend/database is unavailable or restarted.
NFR5	Maintainability	Verify that the layered architecture (presentation / API / data access) supports isolated changes without regressions.
NFR6	Scalability	Verify that the normalized relational schema supports additional entities/records without integrity loss.
NFR7	Availability	Verify that the Docker health check against /api/schemes correctly reports service availability.
NFR8	Accessibility	Verify semantic HTML structure and keyboard focus-visible styling for users with limited technical experience.
NFR9	Data Integrity	Verify foreign-key constraints and unique indexes prevent orphaned or duplicate records.
NFR10	Responsiveness	Verify that layout adapts correctly across common device viewport widths.

6. User Roles and Access Testing

The system defines three distinct roles, as documented in the previous technical report. Access-control testing must additionally confirm that a user authenticated under one role cannot access or perform the functions reserved for another role — this is verified in Section 7 (Scenario TS-13) and Section 9 (Test Cases TC-08, TC-37, TC-41, TC-42, TC-43).

Role	Main Functions	Access That Must Be Tested
Women Entrepreneur	Register, login, manage personal & business profile, browse/search schemes, apply for funding, track applications, enrol in training, register for networking events, request mentorship, view announcements & notifications.	Must access only /entrepreneur/* routes and own records; must be blocked from /officer/* and /mentor/* routes and from other entrepreneurs' data.
Certified Mentor	Login, manage profile, view/accept/reject mentorship requests, conduct sessions, record feedback.	Must access only /mentor/* routes and assigned mentorship requests/sessions; must be blocked from /officer/* routes and from other mentors' request queues.
Government Officer	Login, verify business profiles, manage schemes/funding/training/networking/announcements, review funding applications, manage user accounts, generate reports.	Must access only /officer/* routes; must be blocked from entrepreneur-only or mentor-only actions such as submitting a funding application or accepting a mentorship request.

7. Test Scenario Identification

Eighteen test scenarios (TS-01 to TS-18) were identified from the functional and non-functional requirements above, covering every module and cross-cutting concern of the portal.

Scenario ID	Module	Scenario Description	Requirement ID
TS-01	Authentication	Registration of a new Women Entrepreneur / Certified Mentor account with valid and invalid data.	FR1
TS-02	Authentication	Login with valid and invalid credentials, and session logout.	FR2
TS-03	Business Profile	Creation, update and validation of entrepreneur personal and business (Udyam) profile data.	FR3, FR4
TS-04	Business Verification	Officer approval/rejection of a submitted business profile, including rejection-reason enforcement.	FR19
TS-05	Schemes	Viewing and keyword-searching the government schemes catalogue.	FR5, FR6
TS-06	Schemes	Sector-based filtering of schemes and funding opportunities, including no-match cases.	FR6, FR7
TS-07	Funding	Multi-step funding application submission, validation, amount-limit guard and status tracking.	FR7, FR8, FR9
TS-08	Training	Enrolment in training workshops, duplicate-enrolment guard and capacity guard.	FR10, FR11
TS-09	Networking	Registration for networking events, duplicate-registration guard and capacity guard.	FR15, FR16
TS-10	Mentorship	Mentorship request creation, mentor acceptance/rejection and session feedback.	FR12, FR13, FR14
TS-11	Dashboard	Rendering of the Entrepreneur dashboard with correct statistics and recommendations.	FR21
TS-12	Dashboard	Rendering of the Mentor and Government Officer dashboards with correct statistics.	FR21
TS-13	Authorization	Role-based access control preventing one role from performing another role's actions.	NFR1
TS-14	Integration	Correct end-to-end communication across React frontend, Express REST API and SQLite data layer.	NFR5, NFR9
TS-15	Security	Authentication/authorization security, including expired sessions and direct API access.	NFR1
TS-16	Usability	Clarity of validation messages, navigation and form design.	NFR3, NFR8
TS-17	Reliability	System behaviour under backend/database unavailability and recovery.	NFR4, NFR7

Scenario ID	Module	Scenario Description	Requirement ID
TS-18	Exceptional Handling	Handling of malformed input, request failures and other exceptional/error conditions.	NFR4, NFR9

8. Test Design Techniques

Four complementary test design techniques were applied to derive representative, high-value test cases while keeping the overall suite manageable: Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing.

8.1 Equivalence Partitioning

Equivalence classes were identified for key input fields used across registration, business profile and funding modules.

Field	Valid Partition	Invalid Partition(s)
Email	Well-formed address, unique in system (e.g. anita@example.com)	Malformed address (no @/domain); address already registered (duplicate)
Password	Non-empty string meeting minimum length/complexity policy	Empty string; string below minimum length
Employee Count	Positive whole number (e.g. 1–500, representative)	Negative number; zero (if disallowed); non-numeric value
Turnover	Positive numeric value	Negative value; non-numeric text; empty value
User Role	One of: Women Entrepreneur, Certified Mentor	Any value outside the two permitted registration roles, e.g. 'officer' (officer accounts are provisioned separately, not self-registered)
Required Text Fields (Name, Udyam No., Sector)	Non-empty, within expected length	Empty / whitespace-only value

8.2 Boundary Value Analysis

Boundary Value Analysis was applied to the numerical fields of the business profile and the funding-amount limit. The previous technical report does not publish exact minimum/maximum limits for employee count or turnover in its requirements documentation; the values below are therefore representative test boundaries and must be aligned with the actual validation rules implemented in the code during execution.

Field	Min	Min+1	Normal	Max-1	Max	Below Min (Invalid)	Above Max (Invalid)
Employee Count (representative)	1	2	50	499	500	0 / -1	501
Turnover (representative, INR)	1	2	15,00,000	99,99,999	1,00,00,000	0 / -1	1,00,00,001
Funding Amount (per opportunity max)	1	2	50,000	max-1	max (opportunity-defined)	0 / -1	max+1

8.3 Decision Table Testing

A decision table was constructed for funding-application eligibility, combining business-verification status, opportunity eligibility, and application-data validity.

Rule	Business Verified?	Opportunity Eligibility Satisfied?	Application Data Valid?	Resulting Action
R1	Yes	Yes	Yes	Application accepted and moved to 'Submitted' for officer review (TC-23)

Rule	Business Verified?	Opportunity Eligibility Satisfied?	Application Data Valid?	Resulting Action
R2	No	Yes	Yes	Application blocked with eligibility error (TC-25)
R3	Yes	No	Yes	Application blocked; entrepreneur does not meet sector/turnover eligibility for this opportunity
R4	Yes	Yes	No	Application blocked at the invalid step with a field-level validation error (TC-24, TC-26)
R5	No	No	No	Application blocked; first-failing condition (verification) is reported

8.4 State Transition Testing

State transition testing was applied to the three key stateful entities in the system: the Business Profile, the Funding Application, and the Mentorship Request. This technique is appropriate because each entity's behaviour and the actions available to different roles depend on its current state, and invalid transitions (e.g. re-approving an already verified profile) must be explicitly guarded against, as documented in the previous technical report's issue log.

Object	Current State	Event	Next State
Business Profile	(none)	Entrepreneur submits business form	Pending
Business Profile	Pending	Officer approves	Verified
Business Profile	Pending	Officer rejects (with mandatory remarks)	Rejected
Business Profile	Verified	Officer attempts re-approval	Verified (no change / blocked, TC-17)
Funding Application	(none)	Entrepreneur completes multi-step form	Submitted
Funding Application	Submitted	Officer reviews	Under Review
Funding Application	Under Review	Officer approves	Approved
Funding Application	Under Review	Officer rejects	Rejected
Funding Application	Submitted / Under Review	Entrepreneur withdraws	Withdrawn
Mentorship Request	(none)	Entrepreneur sends request	Pending
Mentorship Request	Pending	Mentor accepts	Accepted
Mentorship Request	Pending	Mentor rejects	Rejected

Note: the Business Profile is created directly in the 'Pending' state (no separate 'Draft' persistence step was found in the previous technical report), which differs from the generic Draft → Pending → Verified/Rejected pattern initially assumed; the table above reflects the workflow as documented.

9. Detailed Test Cases

A total of 53 detailed test cases (TC-01 to TC-53) were designed, covering registration, login, business profile, government verification, schemes, funding, training, networking, mentorship, dashboards, security, exceptional conditions, usability, announcements, notifications, reports and user management. Actual Result and Pass/Fail Status are recorded as “Not Executed” for every test case, since formal execution of this test suite has not yet been carried out for this assignment; these fields must be updated to Pass or Fail once execution is performed.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	Valid Entrepreneur Registration	Navigate to /register; select role 'Women Entrepreneur'; fill all	Name: Anita Sharma; Email: anita.sharma@example.com; Password: Passw0rd!23; Role: Women Entrepreneur	Account is created, JWT token is issued, user is auto-logged-in and	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		mandatory fields with valid data; submit form.		redirected to the entrepreneur dashboard.		
TC-02	Registration - Invalid Email Format	On /register, enter an incorrectly formatted email address and submit.	Email: anita.sharma@@com	Form displays an email-format validation error; account is not created.	Not Executed	Not Executed
TC-03	Registration - Duplicate Email	Attempt to register using an email address that already exists in the users table.	Email: anita.sharma@example.com (existing)	API returns a duplicate-account error (HTTP 400/409); registration is blocked.	Not Executed	Not Executed
TC-04	Registration - Missing Mandatory Field	Submit the registration form leaving the Password field blank.	Password: (empty)	Form displays a 'required field' validation error; registration is blocked.	Not Executed	Not Executed
TC-05	Valid Login	Navigate to /login; enter valid credentials of a registered user; submit.	Email: anita.sharma@example.com; Password: Passw0rd!23	JWT token is stored, user session is created and browser redirects to the role-specific dashboard.	Not Executed	Not Executed
TC-06	Login - Invalid Password	Enter a valid email with an incorrect password.	Email: anita.sharma@example.com; Password: wrongPass1	Login is rejected with an 'invalid credentials' message; no token is issued.	Not Executed	Not Executed
TC-07	Login - Invalid Username	Enter an email that does not exist in the system.	Email: notregistered@example.com; Password: Passw0rd!23	Login is rejected with an 'invalid credentials' message; no token is issued.	Not Executed	Not Executed
TC-08	Login - Unauthorized/Blocked Account Access	Attempt login using an account whose status has been set to inactive/blocked by an officer.	Email of a blocked account; correct password	API rejects login with an appropriate authorization error; user session is not created.	Not Executed	Not Executed
TC-09	Valid Business Profile Creation	Login as entrepreneur; navigate to /entrepreneur/business; complete Udyam number, sector, turnover and employee count; submit.	Udyam No.: UDYAM-TN-12-0012345; Sector: Textiles; Turnover: 1500000; Employees: 8	Business record is created with status 'Pending'; success confirmation is shown.	Not Executed	Not Executed
TC-10	Business Profile - Missing Required Field	Submit the business form without entering the Udyam registration number.	Udyam No.: (empty)	Form displays a required-field validation error; record is not created.	Not Executed	Not Executed
TC-11	Business Profile - Boundary Employee Count	Submit the business form with the minimum representative employee count value.	Employees: 1 (minimum boundary)	Business record is accepted, since 1 is within the valid representative range.	Not Executed	Not Executed
TC-12	Business Profile - Invalid Employee Count	Submit the business form with a negative employee count.	Employees: -5	Form/API rejects the value with a validation error; record is not created.	Not Executed	Not Executed
TC-13	Business Profile - Invalid Turnover	Submit the business form with a non-numeric turnover value.	Turnover: 'abc'	Form/API rejects the value with a validation error; record is not created.	Not Executed	Not Executed
TC-14	Submit Business Profile for Verification	After saving a complete business profile, trigger the submit-for-verification action.	Complete valid business profile	Business status changes from 'Pending' to a state visible in the officer's verification queue.	Not Executed	Not Executed
TC-15	Officer Approves Business Profile	Login as Government Officer; open /officer/entrepreneurs; select a Pending business; approve it.	Business ID of a Pending record	Business status changes to 'Verified'; entrepreneur receives a notification.	Not Executed	Not Executed
TC-16	Officer Rejects Business Profile Without Remarks	As officer, attempt to reject a Pending business without entering a rejection reason.	Rejection remarks: (empty)	API blocks the rejection with a validation error requiring mandatory	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				remarks; status remains unchanged.		
TC-17	Invalid State Transition - Re-verify Verified Profile	As officer, attempt to approve a business that is already in the 'Verified' state.	Business ID of an already Verified record	System rejects/ignores the duplicate transition and preserves the current 'Verified' state without corrupting history.	Not Executed	Not Executed
TC-18	View Government Schemes	Navigate to /entrepreneur/schemes or /public-schemes as an entrepreneur.	N/A	List of active government schemes is rendered correctly.	Not Executed	Not Executed
TC-19	Search Schemes by Keyword	On the schemes page, enter a keyword matching an existing scheme name.	Keyword: 'Startup'	Schemes matching the keyword are displayed; non-matching schemes are excluded.	Not Executed	Not Executed
TC-20	Filter Schemes by Sector	On the schemes page, apply the sector filter for 'Textiles'.	Sector filter: Textiles	Only schemes tagged with the Textiles sector are displayed.	Not Executed	Not Executed
TC-21	No Matching Schemes	Search using a keyword that matches no scheme in the catalogue.	Keyword: 'zzzzzz-nonexistent'	An empty-state message is displayed; no application error occurs.	Not Executed	Not Executed
TC-22	Sector-Based Recommended Schemes	Login as an entrepreneur whose business sector is set; open the entrepreneur dashboard.	Entrepreneur sector: Textiles	Dashboard displays schemes/funding/mentors recommended for the Textiles sector.	Not Executed	Not Executed
TC-23	Valid Funding Application	As a verified entrepreneur, complete the multi-step funding application form with valid data within the programme limit.	Requested amount: 50,000 (within max limit); all mandatory fields completed	Application record is created with status 'Submitted'; confirmation is shown.	Not Executed	Not Executed
TC-24	Funding Application - Missing Information	Attempt to submit the funding application with a mandatory step field left blank.	Business purpose: (empty)	Step is blocked with a validation warning; application is not submitted.	Not Executed	Not Executed
TC-25	Funding Application - Ineligible (Unverified) Entrepreneur	Attempt to submit a funding application while the entrepreneur's business is still in 'Pending' verification status.	Business status: Pending	API blocks the application with an eligibility error referencing unverified business status.	Not Executed	Not Executed
TC-26	Funding Application - Amount Exceeds Maximum Limit	Enter a requested grant amount greater than the funding opportunity's maximum limit.	Requested amount: 5,000,000 (exceeds max limit of 500,000)	Step is blocked with a bounds-check warning toast; application is not submitted.	Not Executed	Not Executed
TC-27	Funding Application - Status Tracking	Open /entrepreneur/applications for a submitted application and review the progress timeline.	Existing Submitted application	Visual step timeline correctly displays Submitted / Review / Sanction stages with officer remarks.	Not Executed	Not Executed
TC-28	Valid Training Enrollment	As entrepreneur, open a training workshop with available seats and enrol.	Workshop with capacity 20, current enrolment 5	Enrollment record is saved; seat count increments by 1; confirmation shown.	Not Executed	Not Executed
TC-29	Training - Duplicate Enrollment	Attempt to enrol again in a workshop the entrepreneur is already registered for.	Same workshop ID as an existing enrolment	API blocks duplicate enrolment using the unique (training_id, entrepreneur_id) index; error message shown.	Not Executed	Not Executed
TC-30	Training - Full/Closed Workshop	Attempt to enrol in a workshop where enrolment count equals capacity.	Workshop capacity: 20, current enrolment: 20	Registration is blocked; a 'Workshop Full' badge/message is displayed.	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-31	Valid Networking Event Registration	As entrepreneur, register for an upcoming networking event.	Event with available capacity	Digital event pass with a unique code is generated; confirmation shown.	Not Executed	Not Executed
TC-32	Networking - Duplicate Registration	Attempt to register again for an event the entrepreneur already registered for.	Same event ID	API blocks duplicate registration; error message shown.	Not Executed	Not Executed
TC-33	Networking - Full/Closed Event	Attempt to register for an event that has reached maximum capacity.	Event capacity reached	Registration is blocked with a 'Event Full' message.	Not Executed	Not Executed
TC-34	Send Mentorship Request	As entrepreneur, select a mentor from the directory and send a mentorship request.	Mentor ID; request message	Request record is created with status 'Pending' in the mentor's queue.	Not Executed	Not Executed
TC-35	Mentor Accepts Mentorship Request	As Certified Mentor, open /mentor/sessions and accept a Pending request.	Pending request ID	Request status changes to 'Accepted'; a session with a meeting link is created.	Not Executed	Not Executed
TC-36	Mentor Rejects Mentorship Request	As Certified Mentor, reject a Pending mentorship request.	Pending request ID	Request status changes to 'Rejected'; entrepreneur is notified.	Not Executed	Not Executed
TC-37	Unauthorized Mentorship Access	As an entrepreneur, attempt to directly call the mentor-only 'accept request' API endpoint.	Direct API call to mentor endpoint using entrepreneur token	API responds with HTTP 403 Forbidden; request status is unchanged.	Not Executed	Not Executed
TC-38	Entrepreneur Dashboard Load	Login as entrepreneur and open the dashboard.	Valid entrepreneur session	Dashboard renders statistics, charts and recommendations correctly with no console errors.	Not Executed	Not Executed
TC-39	Mentor Dashboard Load	Login as certified mentor and open the mentor dashboard.	Valid mentor session	Dashboard renders pending requests, active sessions and statistics correctly.	Not Executed	Not Executed
TC-40	Officer Dashboard Load	Login as government officer and open the officer dashboard.	Valid officer session	Dashboard renders pending verifications, application counts and charts correctly.	Not Executed	Not Executed
TC-41	Unauthorized Role Access (UI Guard)	As an entrepreneur, attempt to navigate directly to an /officer/* route via URL.	URL: /officer/entrepreneurs	User is blocked and redirected away from the officer route.	Not Executed	Not Executed
TC-42	Expired/Invalid Session Token	Send an API request using an expired or tampered JWT token.	Malformed/expired JWT	API responds with HTTP 401 Unauthorized; user is redirected to login.	Not Executed	Not Executed
TC-43	Protected API Access Without Token	Send a request to a protected API endpoint with no Authorization header.	No token supplied	API responds with HTTP 401 Unauthorized.	Not Executed	Not Executed
TC-44	Backend Unavailable	Stop the backend/API container and attempt to load the entrepreneur dashboard.	Backend service stopped	Frontend displays a graceful error/loading-failure message instead of crashing.	Not Executed	Not Executed
TC-45	Database Failure	Simulate an SQLite (sql.js) write failure during a business-profile save operation.	Simulated DB write error	API returns a controlled error response; no partial/corrupt record is persisted.	Not Executed	Not Executed
TC-46	API Request Failure/Timeout	Simulate a network timeout while submitting a funding application.	Simulated network timeout	Frontend shows a retry/error message; application is not partially saved.	Not Executed	Not Executed
TC-47	Invalid/Malformed Input	Submit a malformed JSON payload directly to the registration API endpoint.	Malformed JSON body	API responds with HTTP 400 Bad Request and a descriptive error; no record is created.	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-48	Validation Message Clarity	Trigger several form validation errors across registration, business profile and funding forms.	Various invalid inputs	Validation messages are field-specific, clear and in plain language.	Not Executed	Not Executed
TC-49	Navigation and Form Clarity	Navigate through the entrepreneur, mentor and officer flows end-to-end.	N/A	Navigation is consistent, forms are logically grouped, and role-appropriate menu items are shown.	Not Executed	Not Executed
TC-50	View / Publish Ministry Announcement	As officer, publish a new announcement; as entrepreneur, view the announcements feed.	Announcement title and body	Announcement appears immediately in the entrepreneur's announcement feed.	Not Executed	Not Executed
TC-51	Receive In-App Notification	Trigger a status-changing event (e.g., business verification) and check the notification drawer.	Business verification event	A real-time notification entry appears in the navbar notification drawer.	Not Executed	Not Executed
TC-52	Generate and Export Officer Report (CSV)	As officer, open /officer/reports, apply a filter and export the results.	Filter: date range + module	Filtered report is generated and downloaded as a valid CSV file.	Not Executed	Not Executed
TC-53	Officer Manages User Account Status	As officer, open /officer/users and change an entrepreneur account's status to inactive.	Target user ID	Account status is updated; the affected user is unable to log in while inactive.	Not Executed	Not Executed

10. Test Case Classification

Each of the 53 test cases was classified into exactly one of the following categories, verified against the actual test case list in Section 9.

Category	Count	Test Case IDs
Normal / Positive	23	TC-01, TC-05, TC-09, TC-14, TC-15, TC-18, TC-19, TC-20, TC-22, TC-23, TC-27, TC-28, TC-31, TC-34, TC-35, TC-36, TC-38, TC-39, TC-40, TC-50, TC-51, TC-52, TC-53
Boundary	5	TC-11, TC-21, TC-26, TC-30, TC-33
Invalid / Negative	13	TC-02, TC-03, TC-04, TC-06, TC-07, TC-10, TC-12, TC-13, TC-16, TC-24, TC-25, TC-29, TC-32
Exceptional	5	TC-17, TC-44, TC-45, TC-46, TC-47
Security	5	TC-08, TC-37, TC-41, TC-42, TC-43
Usability	2	TC-48, TC-49

Total classified = 53, matching the total number of designed test cases (53).

11. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) below maps every functional and non-functional requirement identified in Sections 4 and 5 to its corresponding test scenario(s) and test case(s), ensuring no requirement is left unmapped.

11.1 Functional Requirements RTM

Req. ID	Requirement	Scenario ID(s)	Test Case ID(s)	Coverage
FR1	Women Entrepreneur Registration	TS-01	TC-01, TC-02, TC-03, TC-04	Full
FR2	Secure Authentication	TS-02	TC-05, TC-06, TC-07, TC-08	Full
FR3	Entrepreneur Profile Management	TS-03	TC-09, TC-10	Full

Req. ID	Requirement	Scenario ID(s)	Test Case ID(s)	Coverage
FR4	Business Information & Udyam Registration	TS-03	TC-09, TC-11, TC-12, TC-13, TC-14	Full
FR5	Government Schemes Catalogue	TS-05	TC-18	Full
FR6	Scheme Search & Filter	TS-05, TS-06	TC-19, TC-20, TC-21	Full
FR7	Funding Opportunities	TS-07	TC-23	Full
FR8	Funding Applications	TS-07	TC-23, TC-24, TC-25, TC-26	Full
FR9	Application Progress Tracking	TS-07	TC-27	Full
FR10	Training Programme Management	TS-08	TC-28	Full
FR11	Training Enrollment	TS-08	TC-28, TC-29, TC-30	Full
FR12	Mentor Directory	TS-10	TC-34	Full
FR13	Mentorship Requests	TS-10	TC-34, TC-35, TC-36, TC-37	Full
FR14	Mentorship Sessions	TS-10	TC-35, TC-36	Full
FR15	Networking Events	TS-09	TC-31	Full
FR16	Networking Registration	TS-09	TC-31, TC-32, TC-33	Full
FR17	Announcement Management	TS-16	TC-50	Full
FR18	In-App Notifications	TS-11, TS-12	TC-51	Full
FR19	Entrepreneur Verification	TS-04	TC-15, TC-16, TC-17	Full
FR20	Reports & Analytics	TS-12	TC-52	Full
FR21	Dashboards & Statistics	TS-11, TS-12	TC-38, TC-39, TC-40	Full
FR22	User Management	TS-12	TC-53	Full

11.2 Non-Functional Requirements RTM

Req. ID	Requirement	Scenario ID(s)	Test Case ID(s) / Verification Method	Coverage
NFR1	Security	TS-13, TS-15	TC-08, TC-37, TC-41, TC-42, TC-43	Full
NFR2	Performance	TS-14	(observed indirectly during TC-23, TC-38)	Partial - recommend dedicated load/performance testing
NFR3	Usability	TS-16	TC-48, TC-49	Full
NFR4	Reliability	TS-17	TC-44, TC-45, TC-46	Full
NFR5	Maintainability	TS-14	(verified by architecture/code review, not test execution)	Partial - design-level verification only
NFR6	Scalability	TS-14	(verified by schema review, not test execution)	Partial - design-level verification only
NFR7	Availability	TS-17	TC-44	Full
NFR8	Accessibility	TS-16	TC-49	Partial - manual accessibility audit recommended
NFR9	Data Integrity	TS-14, TS-18	TC-03, TC-29, TC-32, TC-45, TC-47	Full
NFR10	Responsiveness	TS-16	(not covered by functional test cases)	Not Covered - recommend dedicated responsive/viewport testing

12. Test Coverage Analysis

- Requirement coverage: all 22 functional requirements (FR1–FR22) are mapped to at least one test case → $22/22 = 100\%$ functional requirement coverage.
- Non-functional requirement coverage: 6 of 10 NFRs (NFR1, NFR3, NFR4, NFR7, NFR9, and partially NFR2/NFR5/NFR6/NFR8) have direct or design-level verification; NFR10 (Responsiveness) has no dedicated test case → $6/10 = 60\%$ fully covered, $9/10 = 90\%$ at least partially addressed, $1/10 = 10\%$ not covered (flagged for future test design).
- Test scenario coverage: all 18 identified scenarios (TS-01–TS-18) have at least one associated test case → $18/18 = 100\%$ scenario coverage.
- Boundary coverage: Boundary Value Analysis was applied to 3 representative numeric fields (employee count, turnover, funding amount), exercised by 5 dedicated boundary test cases (TC-11, TC-21, TC-26, TC-30, TC-33).
- Negative/invalid testing coverage: 13 of 53 test cases ($\sim 24.5\%$) are dedicated negative/invalid cases, in addition to 5 exceptional-condition cases.
- Security coverage: 5 dedicated security test cases (TC-08, TC-37, TC-41, TC-42, TC-43) verify authentication, session validity, and API-level authorization, supported by role-based tests embedded throughout Sections 6 and 9.
- Role-based access coverage: all three roles (Women Entrepreneur, Certified Mentor, Government Officer) are exercised by positive-path test cases, and cross-role restriction is explicitly verified by TC-37 and TC-41.
- Integration coverage: because the application is a three-tier system (React → Express REST API → SQLite/sql.js), every test case in Section 9 indirectly exercises the full request path; TC-44, TC-45 and TC-46 specifically test resilience when the API/database layer fails.

As formal execution of this test suite has not yet been performed for this assignment, execution/pass-rate coverage cannot yet be reported; the percentages above describe design-time (requirement, scenario and technique) coverage only, not execution coverage.

13. Test Execution Status

Formal execution of this CO4 AT1 test suite is currently pending, consistent with the previous technical report's own acknowledgement that comprehensive, independently verified test execution had not been completed at the time of writing. Accordingly, for every one of the 53 test cases in Section 9:

- Actual Result is recorded as: Not Executed
- Pass/Fail Status is recorded as: Not Executed

Once the test cases are executed against the running application, these two fields must be updated on a case-by-case basis to either Pass (actual result matches expected result) or Fail (actual result deviates from expected result), together with supporting evidence (screenshots, logs, or API responses). No test case in this report has been marked as Pass; this document does not claim that testing has already succeeded.

Appendix A provides reference screenshots of the actual implemented interfaces referred to throughout Section 9; these illustrate the system under test but are not execution evidence and carry no Pass/Fail claim.

14. Security Testing

Security testing for the portal focuses on the authentication and authorization mechanisms already implemented (JWT-based sessions with bcrypt password hashing and role-based middleware, as documented in NFR1).

- Authentication: verify that only correctly hashed/matched credentials produce a valid session (TC-05, TC-06, TC-07).
- Authorization: verify that JWT-protected middleware correctly restricts each API route to its intended role(s) (TC-37, TC-41, TC-42, TC-43).
- Role-based access control: verify that entrepreneurs, mentors and officers cannot invoke one another's protected actions, either through the UI or by direct API calls (TC-37, TC-41).

- Protected routes and protected APIs: verify that unauthenticated or unauthorized requests to /officer/*, /mentor/* and their backing API endpoints are rejected (TC-41, TC-43).
- Invalid sessions: verify that expired or tampered tokens are rejected with HTTP 401 and the client session is cleared (TC-42).
- Input validation: verify that malformed or malicious input to registration, business-profile and funding endpoints is rejected rather than persisted (TC-02, TC-04, TC-12, TC-13, TC-47).
- Sensitive data protection: verify that passwords are never returned in API responses and that bcrypt hashes, not plaintext, are stored (verified via code/architecture review, supported by TC-05–TC-08).
- Unauthorized data access: verify that an entrepreneur cannot retrieve another entrepreneur's business, application, or notification records via the API (recommended as an additional test case if not already covered by the existing suite).

15. Reliability and Exception Testing

- Backend unavailable: verify graceful degradation of the frontend when the Express API is unreachable (TC-44).
- Database unavailable: verify that a simulated sql.js write failure returns a controlled error rather than corrupting data (TC-45).
- Network/API failure: verify that a timed-out request during a multi-step form does not leave a partially saved record (TC-46).
- Duplicate submissions: verify that unique constraints (e.g. duplicate Udyam number, duplicate training enrolment, duplicate event registration) are enforced at the API/database level (TC-03, TC-29, TC-32).
- Data persistence: verify that sql.js correctly synchronizes in-memory changes to disk so that data survives a container restart, as claimed under NFR4.
- Incorrect status transitions: verify that invalid state changes (e.g. re-approving an already Verified business) are rejected or ignored without corrupting the record (TC-17).
- Invalid input: verify that malformed JSON or out-of-range values are rejected with a clear HTTP 400 response rather than causing a server error (TC-47).
- Recovery/error messages: verify that all simulated-failure scenarios above produce a user-facing message rather than a blank screen or unhandled exception.

16. Usability Testing

- Navigation: verify that role-appropriate menu items are shown and that navigation between dashboard, profile, schemes, funding, training, mentorship and events is intuitive (TC-49).
- Form clarity: verify that multi-step forms (registration, business profile, funding application) clearly indicate the current step and required fields (TC-49).
- Validation messages: verify that error messages are field-specific and written in plain language, not technical error codes (TC-48).
- Dashboard readability: verify that statistics and charts on all three dashboards are legible and correctly labelled (TC-38, TC-39, TC-40).
- Search/filter usability: verify that scheme search and sector filtering respond promptly and clearly indicate empty results (TC-19, TC-20, TC-21).
- Error message clarity: verify that blocked actions (duplicate enrolment, capacity full, ineligible application) explain why the action was blocked (TC-25, TC-26, TC-29, TC-30, TC-32, TC-33).
- Accessibility for users with limited technical experience: verify semantic HTML structure and visible focus states support keyboard and screen-reader use, per NFR8 (TC-49).

17. Defect Reporting Format

The format below will be used to log any defect discovered once the test cases in Section 9 are executed. No actual defects are reported here, since execution has not yet taken place (Section 13); this section defines only the reporting template and its intended use.

Defect ID	Test Case ID	Defect Description	Severity	Priority	Status
DEF-XX	TC-XX	<To be filled after execution: concise description of observed vs. expected behaviour>	Critical / High / Medium / Low	P1 / P2 / P3	Open / In Progress / Fixed / Closed / Re-opened

During execution, each failed test case (Status = Fail) will generate a corresponding defect record using the format above, linked back to the originating Test Case ID via the RTM in Section 11, so that defect resolution can be re-verified against the same test case.

18. Overall Test Summary

Category	Number
Functional Test Cases	41
Non-Functional Test Cases	12
Positive Test Cases	23
Negative Test Cases	13
Boundary Test Cases	5
Exceptional Test Cases	5
Security Test Cases	5
Total Test Cases	53

The Functional / Non-Functional split and the Positive / Negative / Boundary / Exceptional / Security split are two independent classification dimensions over the same 53 test cases (the latter set excludes the 2 Usability cases, which are counted only within the Non-Functional row). Each dimension sums independently to the Total of 53.

19. Conclusion

This report has designed a comprehensive, traceable test suite for the Women Entrepreneurship Support Portal (Case Study 33 – SDG 5), covering 22 functional and 10 non-functional requirements through 18 test scenarios and 53 detailed test cases. Four test design techniques — Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing — were applied to systematically derive normal, boundary, invalid, exceptional, security and usability conditions, and every requirement was mapped through a Requirements Traceability Matrix to at least one test case.

Executing this suite will materially improve confidence in the system's correctness (accurate scheme, funding and training workflows), reliability (graceful handling of backend/database failures and duplicate submissions), security (enforced authentication, authorization and role-based access control), usability (clear validation and navigation for a non-technical user base), data integrity (unique-constraint and foreign-key enforcement), and role-based access (verified separation of Entrepreneur, Mentor and Officer capabilities) — collectively constituting the quality assurance foundation required before the portal can be responsibly deployed to support women entrepreneurs.

This work directly addresses CO4: “Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics,” by translating the previously implemented system into a disciplined, requirement-traceable test design ready for execution.

20. References

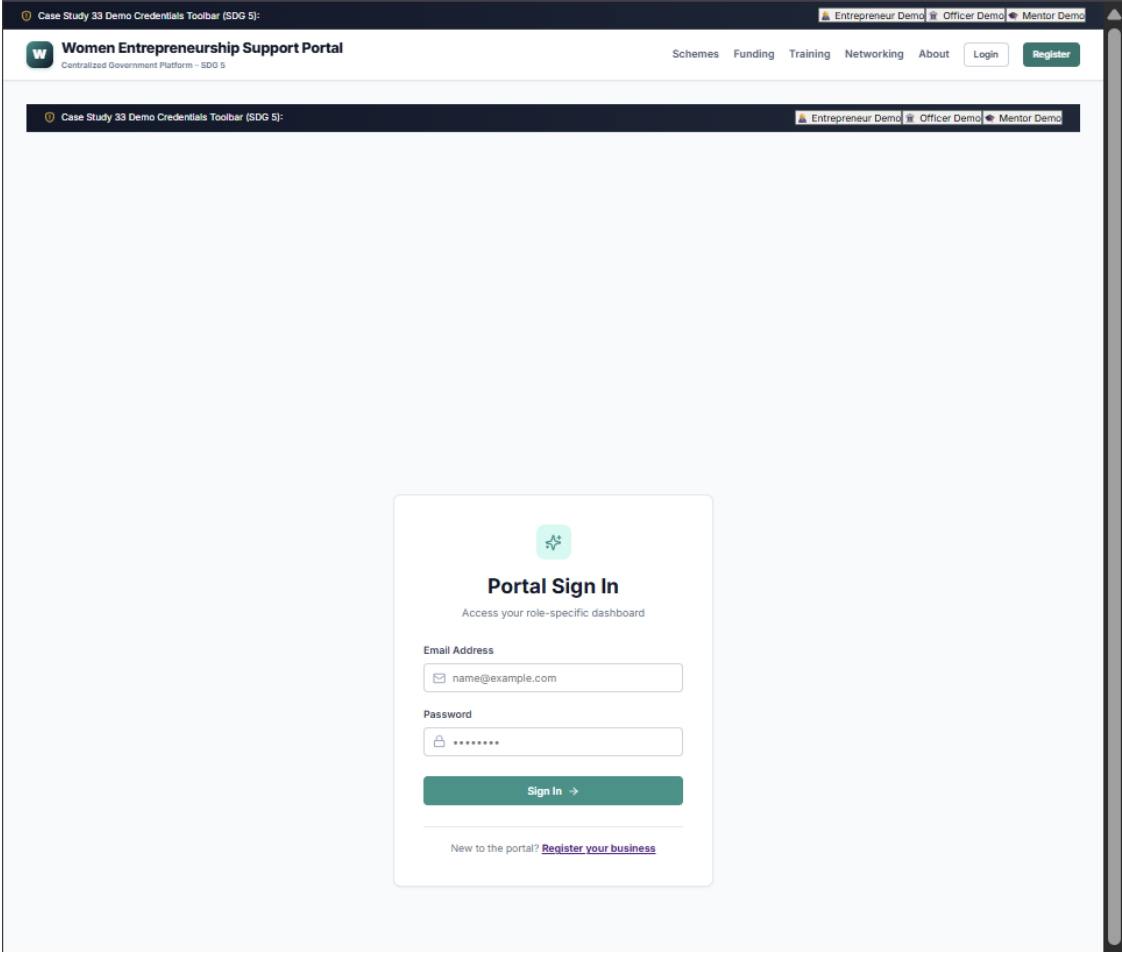
- React Documentation – <https://react.dev>

- Node.js Documentation – <https://nodejs.org/en/docs>
- Express Documentation – <https://expressjs.com>
- MySQL Documentation – <https://dev.mysql.com/doc/> (referenced per assignment prompt; note the previous technical report documents the implemented database engine as SQLite via sql.js)
- Docker Documentation – <https://docs.docker.com>
- Git Documentation – <https://git-scm.com/doc>
- United Nations Sustainable Development Goal 5 – <https://sdgs.un.org/goals/goal5>

Appendix A: System Reference Screenshots

The screenshots below are reproduced from the previously submitted Women Entrepreneurship Support Portal technical/project reports (uploaded as reference material for this assignment). They document the actual implemented interfaces that the test cases in Section 9 exercise, and are provided here purely as visual reference for the reader.

These are screenshots of the running application's user interface, not screenshots of test execution results. As stated in Section 13 (Test Execution Status), this test suite has not yet been formally executed against the application; consequently no Pass/Fail outcome or execution log is claimed or implied by the images below.



The screenshot displays the 'Portal Sign In' interface of the Women Entrepreneurship Support Portal. The page features a dark blue header with the portal's name and navigation links. A central white box contains the sign-in form, which includes fields for email address and password, a 'Sign In' button, and a link for new users to register. The interface is clean and professional, with a light blue background and a subtle grid pattern.

Case Study 33 Demo Credentials Toolbar (SDG 5): Entrepreneur Demo Officer Demo Mentor Demo

Women Entrepreneurship Support Portal
Centralized Government Platform - SDG 5

Schemes Funding Training Networking About Login Register

Case Study 33 Demo Credentials Toolbar (SDG 5): Entrepreneur Demo Officer Demo Mentor Demo

Portal Sign In
Access your role-specific dashboard

Email Address
name@example.com

Password

Sign In →

New to the portal? [Register your business](#)

Fig. A1 - Portal Sign In screen.

Relevant to TC-05, TC-06, TC-07, TC-08 (Login scenarios, TS-02).

Case Study 33 Demo Credentials Toolbar (SDG 5):

Entrepreneur Demo

Officer Demo

Mentor Demo

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

Schemes

Funding

Training

Networking

About

Login

Register

Case Study 33 Demo Credentials Toolbar (SDG 5):

Entrepreneur Demo

Officer Demo

Mentor Demo

Portal Account Registration

Join the Women Entrepreneurship Support Platform (SDG 5)

Women Entrepreneur

Certified Mentor

1. Personal Information

Full Name *

Priya Sharma

Email Address *

priya@example.com

Password *

Phone Number

+91 98765 43210

Date of Birth

dd-mm-yyyy

Highest Education

B.Tech / MBA

City

Jaipur / Bengaluru

State

Rajasthan / Karnataka

Full Address

Street, Industrial Area, Pincode

2. Business Details

Business Name

EcoCraft India

Business Type

Private Limited

Business Sector

Textiles & Handicrafts

Udyam / Reg Number

REG-RJ-2024-9981

Number of Employees

10

Annual Turnover (₹)

2500000

Business Description

Brief summary of your products, market, and impact...

Complete Registration →

localhost:5000/register

Fig. A2 - Portal Account Registration form (role selection, personal & account details).
 Relevant to TC-01, TC-02, TC-03, TC-04 (Registration scenarios, TS-01).

```

kiran@KIRANS_PC MINGW64 ~/gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .env.example
    new file:   .gitignore
    new file:   DBIO_GUIDE.md
    new file:   Dockerfile
    new file:   FINAL_TEST_REPORT.md
    new file:   README.md
    new file:   REQUIREMENTS_TRACEABILITY.md
    new file:   TEST_CASES.md
    new file:   VIVA_QA.md
    new file:   client/index.html
    new file:   client/src/App.jsx
    new file:   client/src/components/DemoToolbar.jsx
    new file:   client/src/components/DocumentViewerModal.jsx
    new file:   client/src/components/Footer.jsx
    new file:   client/src/components/FundingApplicationModal.jsx
    new file:   client/src/components/Modal.jsx
    new file:   client/src/components/Navbar.jsx
    new file:   client/src/components/ProtectRoute.jsx
    new file:   client/src/components/SearchFilterBar.jsx
    new file:   client/src/components/Sidebar.jsx
    new file:   client/src/components/StatCard.jsx
    new file:   client/src/components/StatusBadge.jsx
    new file:   client/src/context/AuthContext.jsx
    new file:   client/src/context/NotificationContext.jsx
    new file:   client/src/context/ToastContext.jsx
    new file:   client/src/index.css
    new file:   client/src/main.jsx
    new file:   client/src/pages/AboutPage.jsx
    new file:   client/src/pages/LandingPage.jsx
    new file:   client/src/pages/LoginPage.jsx
    new file:   client/src/pages/NotFoundPage.jsx
    new file:   client/src/pages/PublicEvents.jsx
    new file:   client/src/pages/PublicFunding.jsx
    new file:   client/src/pages/PublicSchemes.jsx
    new file:   client/src/pages/PublicTraining.jsx
    new file:   client/src/pages/RegisterPage.jsx
    new file:   client/src/pages/UnauthorizedPage.jsx
    new file:   client/src/pages/entrepreneur/AnnouncementsPage.jsx
    new file:   client/src/pages/entrepreneur/BusinessProfilePage.jsx
    new file:   client/src/pages/entrepreneur/EntrepreneurDashboard.jsx
    new file:   client/src/pages/entrepreneur/EntrepreneurFunding.jsx
    new file:   client/src/pages/entrepreneur/EntrepreneurSchemes.jsx
    new file:   client/src/pages/entrepreneur/FundingApplicationModal.jsx
    new file:   client/src/pages/entrepreneur/MentorDirectoryPage.jsx
    new file:   client/src/pages/entrepreneur/MyApplicationsPage.jsx
    new file:   client/src/pages/entrepreneur/MyEventsPage.jsx
    new file:   client/src/pages/entrepreneur/MyMentorshipRequestsPage.jsx
    new file:   client/src/pages/entrepreneur/MyTrainingPage.jsx
    new file:   client/src/pages/entrepreneur/NetworkingEventsPage.jsx
    new file:   client/src/pages/entrepreneur/ProfilePage.jsx
    new file:   client/src/pages/entrepreneur/TrainingProgramsPage.jsx
    new file:   client/src/pages/mentor/MentorDashboard.jsx
    new file:   client/src/pages/mentor/MentorProfilePage.jsx
    new file:   client/src/pages/mentor/MentoredEntrepreneursPage.jsx
    new file:   client/src/pages/mentor/MentoringSessionsPage.jsx
    new file:   client/src/pages/mentor/MentorshipRequestsPage.jsx
    new file:   client/src/pages/officer/AnnouncementManagementPage.jsx
    new file:   client/src/pages/officer/ApplicationManagementPage.jsx
    new file:   client/src/pages/officer/EntrepreneurManagementPage.jsx
    new file:   client/src/pages/officer/FundingManagementPage.jsx
    new file:   client/src/pages/officer/MentorManagementPage.jsx
    new file:   client/src/pages/officer/NetworkingManagementPage.jsx
    new file:   client/src/pages/officer/OfficerDashboard.jsx
    new file:   client/src/pages/officer/ReportsPage.jsx
    new file:   client/src/pages/officer/SchemeManagementPage.jsx
    new file:   client/src/pages/officer/TrainingManagementPage.jsx
    new file:   client/src/pages/officer/UserManagementPage.jsx
    new file:   docker-compose.yml
    new file:   docs/ANNEXURE_A_GIT_DOCKER_ASSIGNMENT.md
    new file:   docs/ANNEXURE_A_SCREENSHOTS.md
    new file:   docs/GIT_DOCKER_COMMANDS.md
    new file:   docs/GIT_GITHUB_FINAL_CHECKLIST.md
    new file:   docs/GIT_SCREENSHOT_COMMANDS.md
    new file:   docs/GIT_SCREENSHOT_GUIDE.md
    new file:   package-lock.json
    new file:   package.json
    new file:   server/db/database.js
    new file:   server/db/initSchema.js

```

Fig. A3 - Entrepreneur Dashboard (logged in as K. Bhavana), showing profile completion, application/training/mentorship statistics and sector-based recommendations.

Relevant to TC-38, TC-22 (Dashboard & recommendation scenarios, TS-11).

```
MINGW64; c:/Users/kiran/Downloads/vaxcare

kiran@KIRANS_PC MINGW64 ~/Downloads/vaxcare (develop)
$ docker compose version
Docker Compose version v5.4.0

kiran@KIRANS_PC MINGW64 ~/Downloads/vaxcare (develop)
$ docker compose config
time="2026-08-18T08:34:21+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\vaxcare\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
name: vaxcare
services:
  backend:
    build:
      context: C:\\Users\\kiran\\Downloads\\vaxcare
      dockerfile: Dockerfile
    container_name: vaxcare_backend
    depends_on:
      db:
        condition: service_healthy
        required: true
    environment:
      DB_HOST: db
      DB_NAME: vaxcare_db
      DB_PASSWORD: vaxcare_pass
      DB_PORT: "3306"
      DB_USER: vaxcare_user
      PORT: "5000"
      SECRET_KEY: vaxcare_docker_secret_key_2026
    networks:
      vaxcare_network: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
    restart: always
  db:
    container_name: vaxcare_mysql
    environment:
      MYSQL_DATABASE: vaxcare_db
      MYSQL_PASSWORD: vaxcare_pass
      MYSQL_ROOT_PASSWORD: vaxcare_root_pass
      MYSQL_USER: vaxcare_user
    healthcheck:
      test:
        - CMD
        - mysqladmin
        - ping
        - -h
        - localhost
        - -u
        - root
        - -pvaxcare_root_pass
      timeout: 5s
      interval: 5s
      retries: 10
```

Fig. A4 - Business Profile & Registration form capturing personal information and business/Udyam details.
Relevant to TC-09 to TC-14 (Business Profile scenarios, TS-03).

Portal Account Registration
Join the Women Entrepreneurship Support Platform (SEDS)

Women Entrepreneur | **Certified Mentor**

1. Personal Information

Full Name *
Email Address *

Password *
Phone Number *

Date of Birth *
Highest Education *

City *
State *

Full Address *

2. Business Details

Business Name *
Business Type *

Business Sector *
Udyam / Reg Number *

Number of Employees *
Annual Turnover (₹) *

Business Description *

Complete Registration →

Already registered? [Sign In Here](#)

Fig. A5 - Government Schemes Directory with search and sector filter.
Relevant to TC-18 to TC-22 (Scheme search/filter scenarios, TS-05, TS-06).

Women Entrepreneurship Support Portal

Welcome back, K Bhavana!
Business: EcoCraft India (Textiles & Handicrafts)

Profile Completion Meter 80% Completed
Complete your profile to access more relevant opportunities. Government Business Verification [Complete Now](#)

0 Active Applications | **0** Approved Grants | **0** Scheduled Training Workshops | **0** Membership Requests

Recommended for You

Recommended Schemes

- Stand Up India Women Enterprise**
Borrowing Scheme
Ministry of PwD, Small and Medium Enterprises
- Mahila Samakhya Yojana for Cottage, Cluster & Handicrafts**
Ministry of Textiles & Handicrafts

[View All Schemes](#)

Recommended Grants & Funding

- Women Green Business Expansion Grant**
Grant: ₹2,000,000
- Early Stage Women Startup Seed Capital**
Grant: ₹5,000,000

[Apply to this Funding](#)

Recommended Advisors & Mentors

- Dr. Savita Rao**
Project Management, Fund & Equity Fundraising, Business Scale-up Strategy
- Radhika Nar**
Export Documentation, Customs Compliance, Global Supply Chain Logistics

[Connect with Mentors](#)

Quick Links

- Government Schemes
- Funding Grants
- Training Workshops
- Networking Events
- About SEDS & Support

Policy & Help

- Privacy Policy
- Terms of Service
- Support & FAQ
- Official Portal Login

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Fig. A6 - Funding & Capital Opportunities listing with amount ranges and deadlines.
Relevant to TC-23 to TC-27 (Funding application scenarios, TS-07).

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

OFFICER

Rajesh Varma

Logout

GOVERNMENT ADMIN WORKSPACE

Rajesh Varma

Dashboard

Entrepreneurs & Verification

Schemes

Funding

Applications

Training

Mentors

Networking

Announcements

Reports & Analytics

User Management

Entrepreneur & Business Verification

Audit registered female founders, inspect Udyam registration credentials, and issue government verification badges

Search by keywords, codes, names...

All Verification Status

All Sector

Reset

FOUNDER NAME	BUSINESS NAME & UDYAM REG	SECTOR	TURNOVER & EMPLOYEES	VERIFICATION STATUS	ACTIONS
THUMMURU JAYA 192572036.simaraj@xaveetha.com 9874563210	eco craft reg-rj-0987	Textiles & Handicrafts	Turnover: ₹2,500,000 10 Employees	Verified	Verify Reject
Meera Reddy meera@womenportal.test +91 97654 32109	UrbanClean Mobility Systems REG-TS-2024-11204	Technology & CleanTech	Turnover: ₹1,500,000 8 Employees	Rejected	Verify Reject
Ananya Verma ananya@womenportal.test +91 98123 45678	BioNutra Organics Pvt Ltd REG-KA-2021-99432	Biotechnology & Healthcare	Turnover: ₹8,200,000 18 Employees	Verified	Verify Reject
Priya Sharma entrepreneur@womenportal.test +91 98765 43210	EcoCraft India Handicrafts REG-RJ-2022-88741	Textiles & Handicrafts	Turnover: ₹4,500,000 24 Employees	Verified	Verify Reject

Fig. A7 - Officer view: Entrepreneur & Business Verification queue with Verify/Reject actions and status badges. Relevant to TC-15, TC-16, TC-17 (Government Verification scenarios, TS-04).

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

OFFICER

Rajesh Varma

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GOVERNMENT ADMIN WORKSPACE

Rajesh Varma

Dashboard

Entrepreneurs & Verification

Schemes

Funding

Applications

Training

Mentors

Networking

Announcements

Reports & Analytics

User Management

Funding Application Review Queue

Evaluate submitted grant applications, review business plans, and issue official approvals or rejections

Search by keywords, codes, names...

All Status Queue

Reset

APP NUMBER	APPLICANT BUSINESS	FUNDING SCHEME	REQUESTED AMOUNT	STATUS	OFFICER REMARKS	ACTIONS
APP-2026-8803	UrbanClean Mobility Systems Owner: Meera Reddy	Tech-Innovation Concessional Micro-Loan	₹500,000	Submitted	None	Docs Evaluate
APP-2026-8802	BioNutra Organics Pvt Ltd Owner: Ananya Verma	Early-Stage Women Startup Seed Capital	₹1,200,000	Under Review	Technical committee is evaluating the patent documentation and laboratory test certificates.	Docs Evaluate
APP-2026-8801	EcoCraft India Handicrafts Owner: Priya Sharma	Women Green Business Expansion Grant Government Grant	₹1,500,000	Approved	Outstanding application. Fully verified environmental compliance and strong community impact.	Docs Evaluate

Logout

Fig. A8 - Officer view: Funding Application Review Queue showing applicant, scheme, requested amount, status and officer remarks.
Relevant to TC-23 to TC-27 and the funding-eligibility Decision Table (Section 8.3).

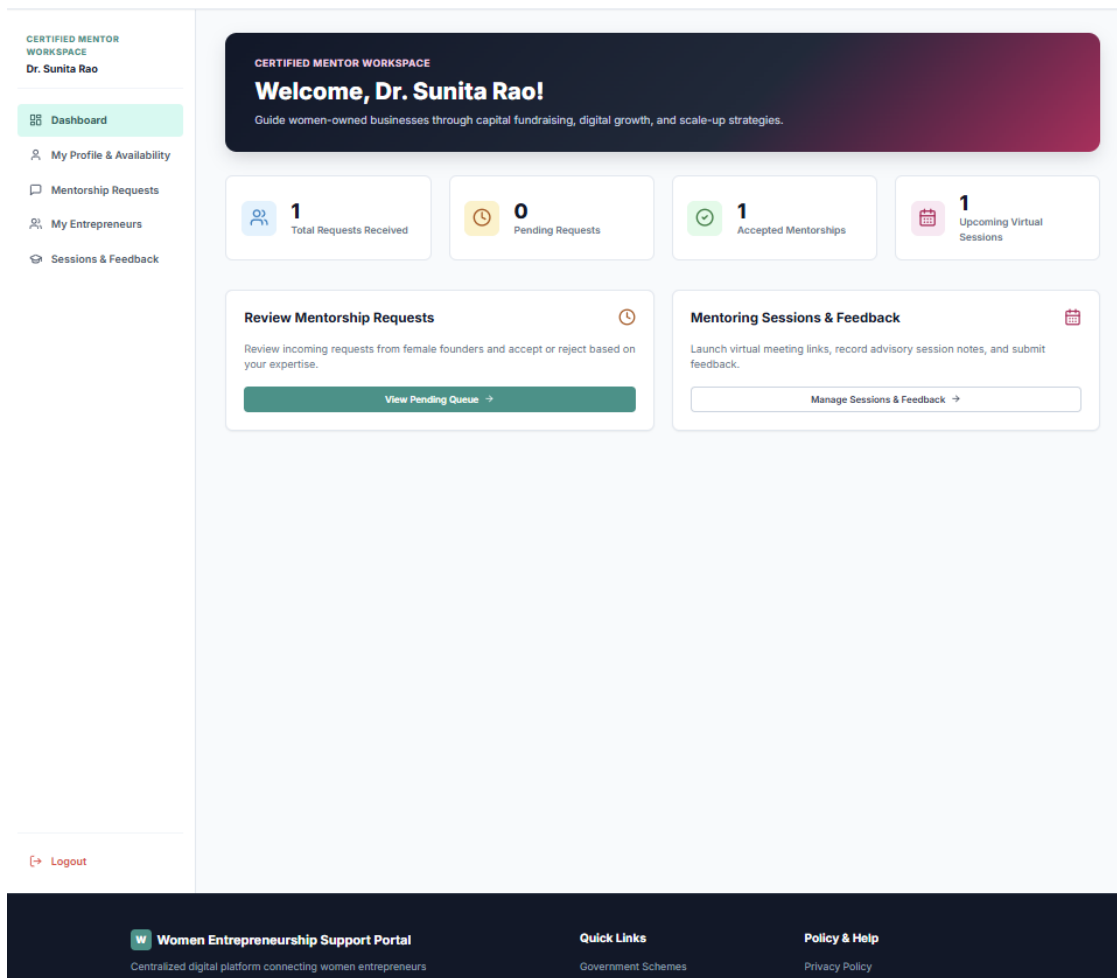


Fig. A9 - Certified Mentor Dashboard showing mentorship requests, accepted mentorships and upcoming sessions.
Relevant to TC-34, TC-35, TC-36, TC-39 (Mentorship & Dashboard scenarios, TS-10, TS-12).

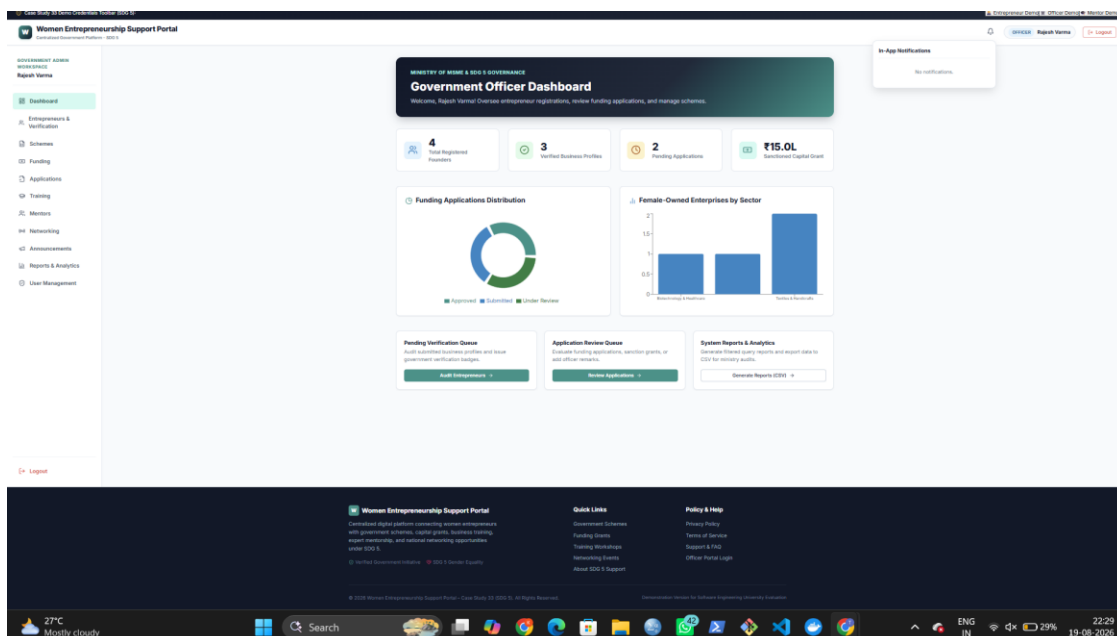


Fig. A10 - Government Officer Dashboard showing registered founders, verified profiles, pending applications, sanctioned capital and sector-wise distribution charts.
Relevant to TC-40 (Officer Dashboard scenario, TS-12).

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Mentor Directory Management

Oversee certified mentors, qualifications, and availability schedules

Mentor Name	Professional Title & Industry	Expertise	Experience	Availability	Status
Dr. Sunita Rao mentor@womenportal.test +91 93456 78901	Senior Partner at Venture Horizon Capital, Ex-Director SIDBI Venture Funds Finance & Venture Capital	Financial Management, Debt & Equity Fundraising, Business Scale-up Strategy	18 Years	Mon & Thu (4:00 PM - 7:00 PM)	Active
Radhika Nair radhika.mentor@womenportal.test +91 95678 90123	Chief Supply Chain Officer at Apex Logistics, Director Export Promotion Council Export Management & Manufacturing	Export Documentation, Customs Compliance, Global Supply Chain Logistics	16 Years	Tue & Fri (3:00 PM - 6:00 PM)	Active
Kavita Krishnan kavita.mentor@womenportal.test +91 84567 89012	Founder & CEO at GrowthVeda Digital, Ex-VP Marketing Flipkart Digital Marketing & E-Commerce	Brand Building, Performance Marketing, Global E-Commerce Export Strategy	12 Years	Wed & Sat (2:00 PM - 5:00 PM)	Active

Fig. A11 - Officer view: Mentor Directory Management listing certified mentors, expertise and availability.
Relevant to FR12 (Mentor Directory) and TC-34 (mentor selection).

```
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:28:24+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                                COMMAND
SERVICE            CREATED        STATUS      PORTS
women_entrepreneurship_portal_app  women-entrepreneurship-portal-portal-app  "docker-entrypoint.s..."
portal-app            19 minutes ago  Up 19 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal>
```

Fig. A12 - Docker Compose service status showing the application container in a healthy state with the API/DB ports bound.
Relevant to TC-44, TC-45 and NFR7 (Availability/Reliability, TS-17).

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete

Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25